



Crave It • Search It • Cook It

A Multimodal Recipe RAG Assistant

Architecture • Multilingual & Multimodal Retrieval • LLM-as-Judge Evaluation

LangChain • FAISS • OpenCLIP • Groq gpt-oss-120b • Gradio

What It Does

A chat-based recipe assistant that understands any-language queries, finds the closest matching recipes by meaning, surfaces visually similar dish photos, and generates a grounded, chef-style answer from real retrieved recipes.



Multilingual by default

Auto-detects & translates 20+ languages before retrieval



Semantic recipe search

FAISS + MiniLM embeddings for meaning-based search



Visual dish matching

CLIP finds visually similar dishes, even across languages



Grounded chat answers

Groq gpt-oss-120b writes from retrieved context, not memory

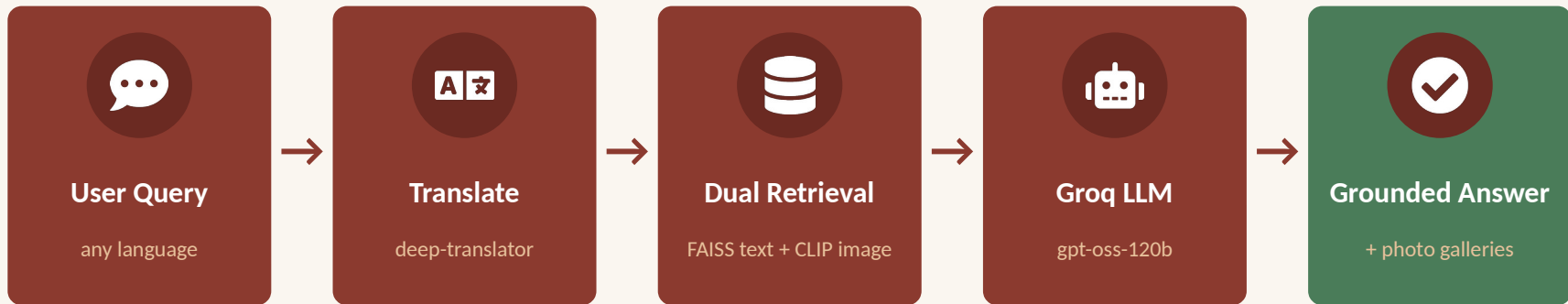


"मसाला दाल"

→ translated → retrieved → answered

One chat box.
Any language.
Real recipes only.

How It Works: RAG Pipeline



The text index (MiniLM embeddings) finds the closest-meaning recipes; the CLIP index finds visually similar dish photos independently — both feed into a single Gradio interface.

Text Index

sentence-transformers/all-MiniLM-L6-v2 → FAISS

Image Index

OpenCLIP ViT-B-32 (openai weights) → FAISS

Tech Stack



Gradio

Chat UI, image galleries, session state



LangChain

Orchestrates retrieval + prompt + LLM chain



FAISS

Vector similarity search, text & image



Sentence-Transformers

MiniLM-L6-v2 text embeddings



OpenCLIP

ViT-B-32 joint text/image embedding space



Groq API

gpt-oss-120b generation, citation tags stripped for display



deep-translator + langdetect

Any-language query support



Custom LLM-judge harness

Faithfulness / relevancy scoring

Evaluating a RAG System: LLM-as-Judge

Standard RAG apps have no ground-truth "correct answer" to check against. A separate LLM judge scores each generated answer against what was actually retrieved.



Faithfulness (1-5)

Is the answer grounded ONLY in retrieved recipe context — no invented ingredients or steps?



Relevancy (1-5)

Does the answer actually address what the user asked for?



Completeness (0/1)

Deterministic check: does the answer include both ingredients AND steps?



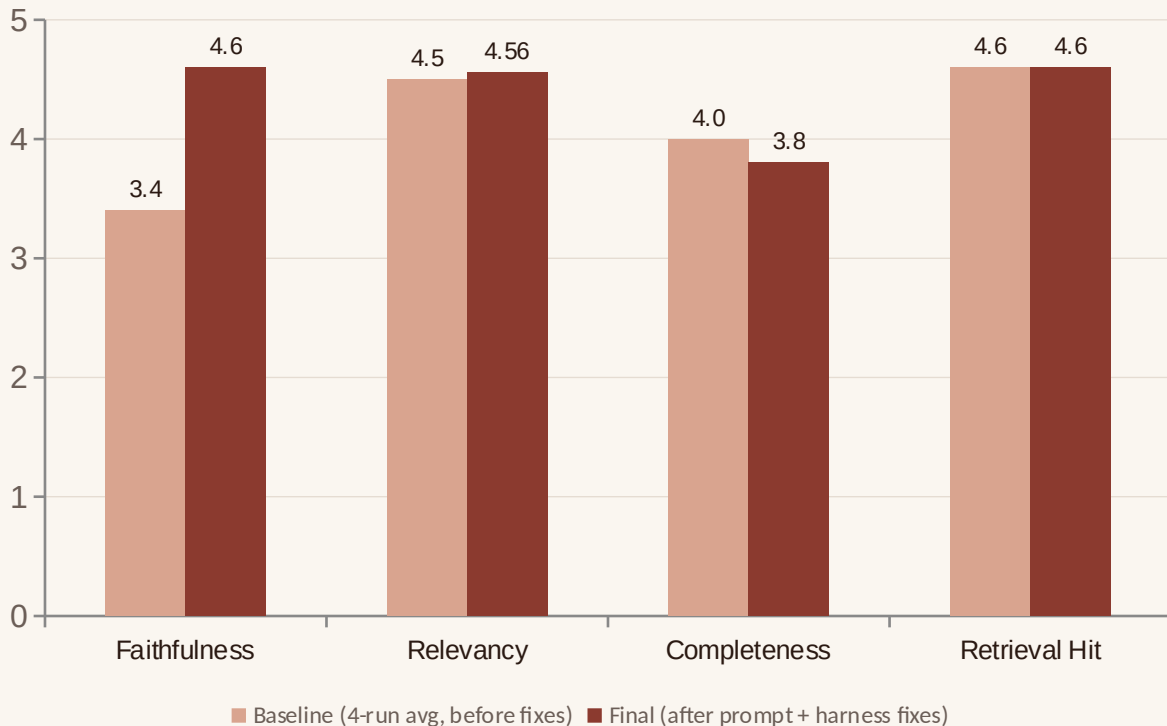
Retrieval Hit Rate

Deterministic keyword check: did retrieval actually surface the right dish?

Key methodology choice: each test case is run once, and the answers are consistent. Truncation was also removed so no information is lost.

Evaluation Results

Averaged across 4 baseline runs, then re-validated after fixes



4.6/5

Avg faithfulness (23 of 25 cases scored 5)

99.2%

Citation accuracy (deterministic check)

92%

Retrieval Hit Rate

0%

Genuine model errors (2 quota 429s)



Before / After: Closing the Faithfulness Gap

Baseline runs showed the model blending its own training-data knowledge into answers for globally-famous dishes, even when it contradicted the retrieved recipe. A citation-forcing prompt rewrite — every claim tagged to its source recipe — closed that gap entirely.

Dish	Before	After
Carbonara	2/5	5/5
Tiramisu	1/5	5/5
Tikka Masala	1/5	5/5
Shrimp Dumplings	2/5	5/5
Naan / Crêpes / Paneer	2-3/5	5/5



7 of 8 cases with 100% verified citations jumped straight to 5/5 once the judge could see all 5 retrieved recipes (not just the first 2) and had enough token budget to finish its verdict.

Engineering Challenges & Fixes



Citation-forcing + verification

Prompt now tags every claim to its source recipe; a separate non-LLM check verifies each tag against real retrieved titles (99%+ accurate)



Judge context truncation

Judge only saw ~2 of 5 retrieved recipes on long comparisons — removed truncation entirely so it sees everything the generator did



Empty judge output

Reasoning models (gpt-oss) can exhaust their token budget on hidden reasoning before writing a verdict — raised `max_tokens`



Groq daily (not per-minute) quota

A 429 that no retry-sleep fixed — turned out to be a 200k-tokens/day cap, confirmed from the raw API error text

Skills Demonstrated

- ✓ RAG system design (dual retrieval: text + image)
- ✓ Multilingual NLP pipelines
- ✓ LLM-as-judge evaluation, cross-checked with deterministic verification
- ✓ Prompt engineering that measurably closed a faithfulness gap (3.41 → 5.0/5)
- ✓ API resilience: distinguishing infra failures (daily quotas) from real model errors
- ✓ Root-cause debugging of eval-harness bias, not just accepting the score

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